FACULTY OF ENGINEERING & TECHNOLOGY
ELECTRICAL AND COMPUTER ENGINEERING DEPARTMENT

Project #1: Optimization Strategies for Local Package Delivery Operations

Team Members:

- Name: Laith Muayad Shhadeh ID: 1241687
- Name: Yousef shanti ID:1221137

Submission Date: May 04, 2025

Dr. Yazan Abu Farha
Dr. Samah Alaydi

Table of contents :

Introduction……..3
Problem Formulation……3
Heuristic Techniques Used……...4
Handling Constraints……4
Parameter Tuning……..5
Test Cases and Observations……6
Conclusion……...10

Table of Figures :

Figure[1].....6
Figure[2].....6
Figure[3].....7
Figure[4].....7
Figure[5].....8
Figure[6].....8
Figure[7].....9
Figure[8].....9
Figure[9].....9

Introduction

Efficient delivery route planning is essential in logistics to minimize operational costs and ensure timely deliveries. This report explores the application of two popular metaheuristic algorithms—Simulated Annealing (SA) and Genetic Algorithm (GA)—to solve the Vehicle Routing Problem (VRP) with capacity constraints and prioritized packages.

Problem Formulation

The goal is to assign a set of packages to a fixed number of delivery vehicles such that:

- No vehicle exceeds its weight capacity.
- Total distance traveled by all vehicles is minimized.
- High-priority packages are considered first.

Each package is defined by:

- A destination represented as a 2D coordinate.
- A weight in kilograms.
- A priority (1 being highest, 5 being lowest).

Each vehicle:

- Has a fixed weight capacity.
- Starts and ends at the origin (0, 0).

The cost of a route is calculated as the total Euclidean distance from the origin through all assigned package locations and back to the origin.

Heuristic Techniques Used

Simulated Annealing (SA)

SA is inspired by the annealing process in metallurgy. It begins with a high "temperature" allowing for more explorative changes in the solution and gradually cools down to narrow in on an optimal solution.

- **Initial Solution**: Packages are assigned in round-robin fashion across vehicles.
- **Neighbor Generation**: Two packages are swapped between two randomly selected vehicles.
- **Acceptance Probability**: A worse solution is accepted with a probability that decreases with temperature.

Genetic Algorithm (GA)

GA mimics biological evolution. A population of solutions evolves over generations via selection, mutation, and reproduction.

- **Initialization**: Random shuffling of packages with round-robin assignment.
- **Selection**: Top-performing solutions (elitism) are carried forward.
- **Mutation**: Swapping packages between vehicles.
- **Reproduction**: Cloning elite or top-ranking individuals.

Handling Constraints

Vehicle Capacity

Packages whose weight exceeds the maximum vehicle capacity are excluded from consideration. This is reported to the user during input.

Priority Handling

All packages are sorted in ascending order of priority (i.e., 1 first) before being assigned in either algorithm. This ensures higher-priority packages are more likely to be assigned early in the process.

Overload Prevention

After each mutation or neighbor generation, a validity check ensures that no vehicle exceeds its capacity. If any vehicle is overloaded, the solution is discarded and not considered.

Parameter Tuning

Simulated Annealing

- **Initial Temperature**: A high temperature of 1000 was used to allow broad exploration early on. Decreasing this value led to faster convergence but higher risk of local minima.
- **Cooling Rate**: Set to 0.95. A lower cooling rate (e.g., 0.90) accelerated the annealing process but often led to suboptimal solutions. A higher rate (e.g., 0.98) improved solution quality but increased computation time.

Genetic Algorithm

- **Population Size**: A size of 50 provided a balance between diversity and computation time. Smaller populations reduced coverage of the solution space.
- **Mutation Rate**: A value of 0.05 maintained diversity in the population. Lower mutation rates caused stagnation, while higher rates disrupted good solutions.
- **Elite Retention**: The top 10 solutions were preserved each generation to ensure the best candidates were not lost.

Test Cases and Observations

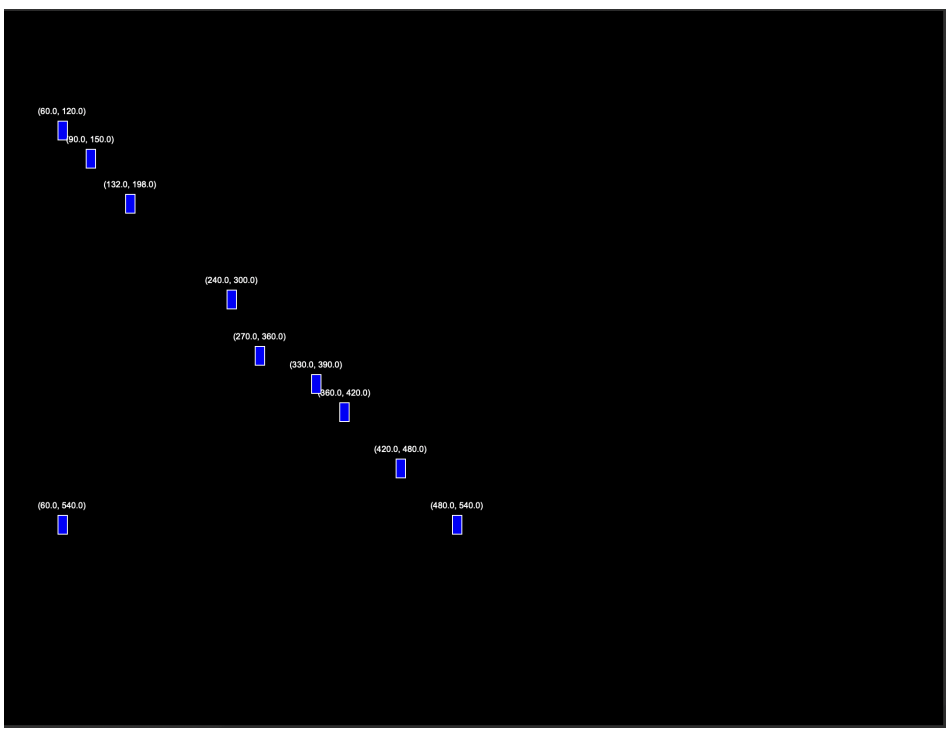
Test Case 1: Feasibility Check

- **Input**: 3 vehicles, capacity 100kg, 10 packages
- **Result**:
-

```
Choose algorithm (1 = Simulated Annealing, 2 = Genetic Algorithm): 1

Vehicle 1 (Total Weight: 75.00 kg, Distance: 281.03 km):
  - Dest: (10.0,90.0) | Weight: 5.0kg | Priority: 1
  - Dest: (80.0,90.0) | Weight: 15.0kg | Priority: 1
  - Dest: (70.0,80.0) | Weight: 25.0kg | Priority: 3
  - Dest: (60.0,70.0) | Weight: 30.0kg | Priority: 2

Vehicle 2 (Total Weight: 45.00 kg, Distance: 171.42 km):
  - Dest: (40.0,50.0) | Weight: 20.0kg | Priority: 3
  - Dest: (55.0,65.0) | Weight: 5.0kg | Priority: 4
  - Dest: (45.0,60.0) | Weight: 20.0kg | Priority: 2

Vehicle 3 (Total Weight: 50.00 kg, Distance: 79.72 km):
  - Dest: (10.0,20.0) | Weight: 30.0kg | Priority: 1
  - Dest: (15.0,25.0) | Weight: 10.0kg | Priority: 2
  - Dest: (22.0,33.0) | Weight: 10.0kg | Priority: 2
```

Fig[1]

-



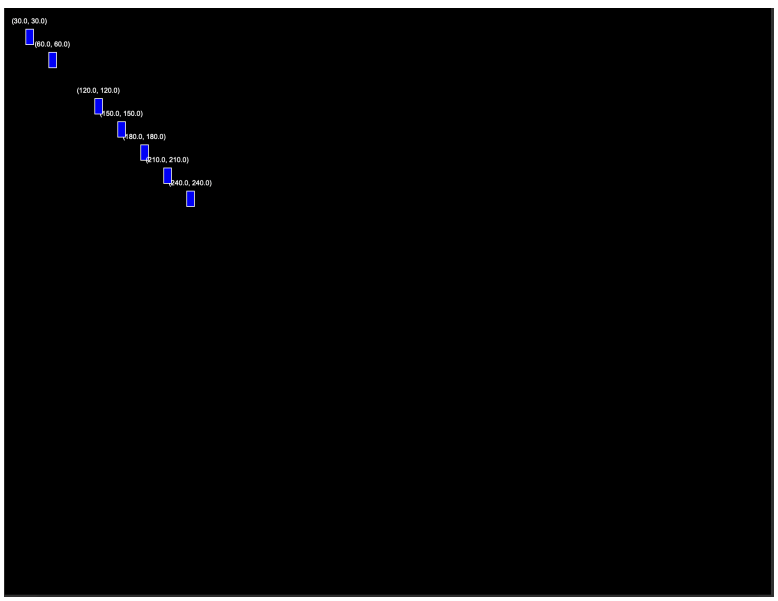
Fig[2]

Test Case 2: Priority Biasing

- **Input**: 2 vehicles, some packages marked as priority 1
- **Result**:
-

```
Choose algorithm (1 = Simulated Annealing, 2 = Genetic Algorithm): 2

Vehicle 1 (Total Weight: 75.00 kg, Distance: 281.03 km):
  - Dest: (10.0,90.0) | Weight: 5.0kg | Priority: 1
  - Dest: (80.0,90.0) | Weight: 15.0kg | Priority: 1
  - Dest: (70.0,80.0) | Weight: 25.0kg | Priority: 3
  - Dest: (60.0,70.0) | Weight: 30.0kg | Priority: 2

Vehicle 2 (Total Weight: 45.00 kg, Distance: 171.54 km):
  - Dest: (55.0,65.0) | Weight: 5.0kg | Priority: 4
  - Dest: (45.0,60.0) | Weight: 20.0kg | Priority: 2
  - Dest: (40.0,50.0) | Weight: 20.0kg | Priority: 3

Vehicle 3 (Total Weight: 50.00 kg, Distance: 79.84 km):
  - Dest: (10.0,20.0) | Weight: 30.0kg | Priority: 1
  - Dest: (22.0,33.0) | Weight: 10.0kg | Priority: 2
  - Dest: (15.0,25.0) | Weight: 10.0kg | Priority: 2
```

Fig[3]

- 

Fig[4]

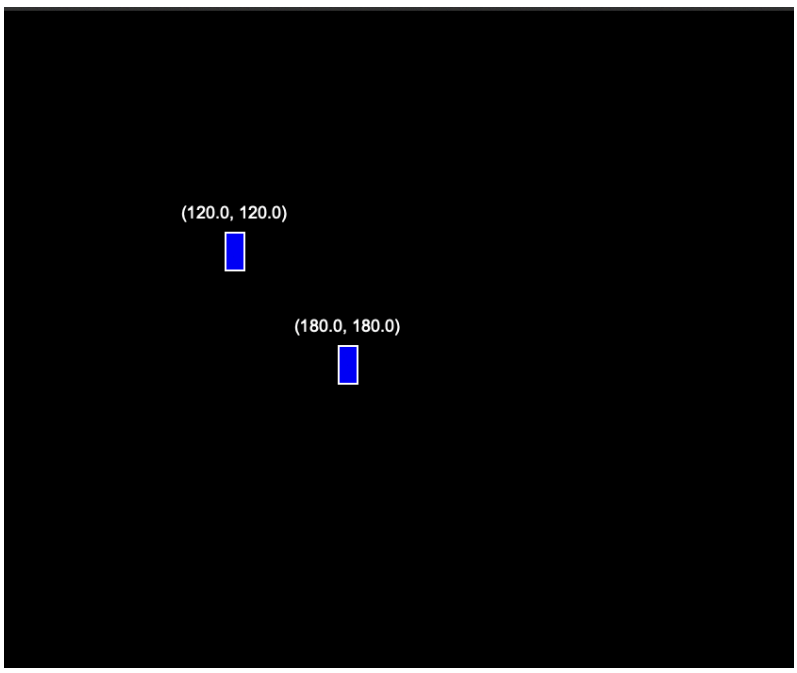
Test Case 3: Overcapacity Packages

- **Input**: One package with 150kg weight, vehicle capacity 100kg
- **Result :**
-

```
Choose algorithm (1 = Simulated Annealing, 2 = Genetic Algorithm): 1
Skipping package at (10.0, 10.0) — weight exceeds capacity.

Vehicle 1 (Total Weight: 90.00 kg, Distance: 56.57 km):
  - Dest: (20.0,20.0) | Weight: 90.0kg | Priority: 1

Vehicle 2 (Total Weight: 50.00 kg, Distance: 84.85 km):
  - Dest: (30.0,30.0) | Weight: 50.0kg | Priority: 2
```

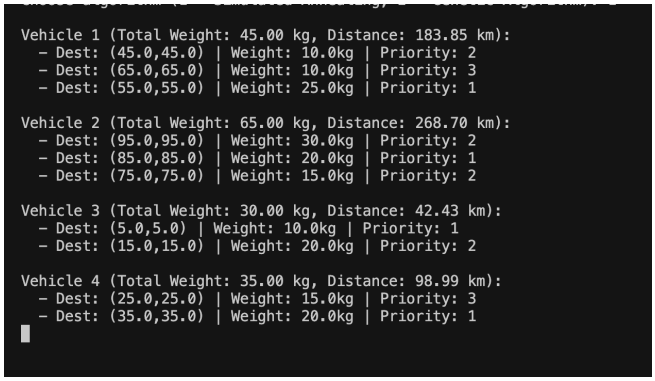
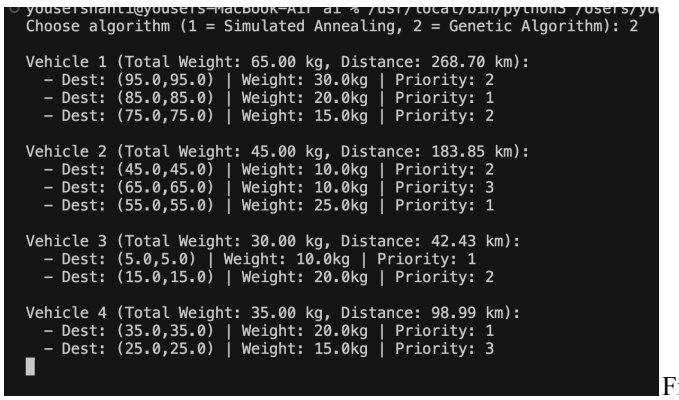
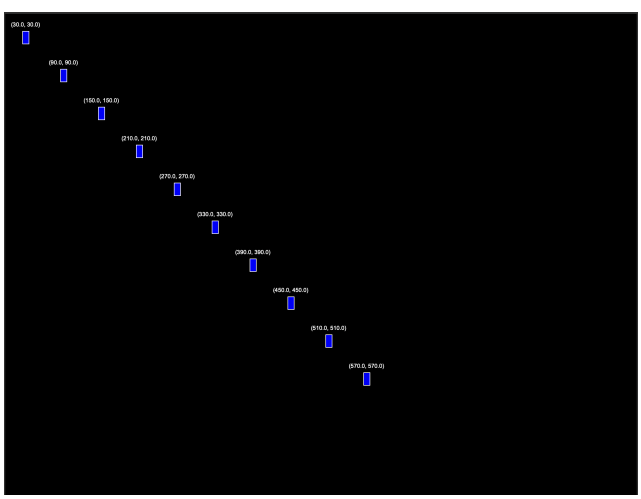
Fig[5]

-



Fig[6]

Test Case 4:

- input : SA generally produced shorter total distances for small datasets. GA scaled better for large input sizes due to population diversity.
- Result :
- 
```
Vehicle 1 (Total Weight: 45.00 kg, Distance: 183.85 km):
  - Dest: (45.0,45.0) | Weight: 10.0kg | Priority: 2
  - Dest: (65.0,65.0) | Weight: 10.0kg | Priority: 3
  - Dest: (55.0,55.0) | Weight: 25.0kg | Priority: 1

Vehicle 2 (Total Weight: 65.00 kg, Distance: 268.70 km):
  - Dest: (95.0,95.0) | Weight: 30.0kg | Priority: 2
  - Dest: (85.0,85.0) | Weight: 20.0kg | Priority: 1
  - Dest: (75.0,75.0) | Weight: 15.0kg | Priority: 2

Vehicle 3 (Total Weight: 30.00 kg, Distance: 42.43 km):
  - Dest: (5.0,5.0) | Weight: 10.0kg | Priority: 1
  - Dest: (15.0,15.0) | Weight: 20.0kg | Priority: 2

Vehicle 4 (Total Weight: 35.00 kg, Distance: 98.99 km):
  - Dest: (25.0,25.0) | Weight: 15.0kg | Priority: 3
  - Dest: (35.0,35.0) | Weight: 20.0kg | Priority: 1
```
Fig[7]
- 
```
Choose algorithm (1 = Simulated Annealing, 2 = Genetic Algorithm): 2

Vehicle 1 (Total Weight: 65.00 kg, Distance: 268.70 km):
  - Dest: (95.0,95.0) | Weight: 30.0kg | Priority: 2
  - Dest: (85.0,85.0) | Weight: 20.0kg | Priority: 1
  - Dest: (75.0,75.0) | Weight: 15.0kg | Priority: 2

Vehicle 2 (Total Weight: 45.00 kg, Distance: 183.85 km):
  - Dest: (45.0,45.0) | Weight: 10.0kg | Priority: 2
  - Dest: (65.0,65.0) | Weight: 10.0kg | Priority: 3
  - Dest: (55.0,55.0) | Weight: 25.0kg | Priority: 1

Vehicle 3 (Total Weight: 30.00 kg, Distance: 42.43 km):
  - Dest: (5.0,5.0) | Weight: 10.0kg | Priority: 1
  - Dest: (15.0,15.0) | Weight: 20.0kg | Priority: 2

Vehicle 4 (Total Weight: 35.00 kg, Distance: 98.99 km):
  - Dest: (35.0,35.0) | Weight: 20.0kg | Priority: 1
  - Dest: (25.0,25.0) | Weight: 15.0kg | Priority: 3
```
Fig[8]
- 

Fig[9]

Conclusion

Both Simulated Annealing and Genetic Algorithm provide viable solutions to the VRP with capacity and priority constraints. Simulated Annealing is simpler and often more intuitive to implement, while Genetic Algorithm offers more flexibility and scalability for large datasets. Prioritizing high-importance packages and enforcing strict capacity checks ensured the reliability and practical relevance of the implemented system.